

# Setting up a Web Server on Raspberry Pi



Raspberry Pi is powerful enough to act as a Web server with complete database and PHP programming support. Here we describe how you can turn your Raspberry Pi into a Web server and host your own website on it

POOJA JUVAL

The primary function of a Web server is to deliver web-pages on the request of clients using the Hypertext Transfer Protocol (HTTP). This means delivery of HTML documents and any additional content that may be included by a document, such as images, style sheets and scripts, as defined by Wikipedia.

Making Raspberry Pi act as a Web server requires a lot of additional software other than the basic operating system. The steps below will guide you through installation of operating system and additional software, change in configuration files, and finally website hosting on Raspberry Pi. Fig. 1 shows the final website where a sample website template taken from the Internet is used and hosted on Raspberry Pi. Designing website is out of the scope of this article.

## Install Raspbian 'Wheezy'

Download the image of the operating system from <http://www.raspberrypi.org/downloads>. After downloading the zip file, extract the image in a suitable folder and then insert a 2GB or 4GB SD card using a card reader to your computer. At this stage, before moving further, determine beyond doubt what identity is given to the SD card by your computer. Any mistake here will lead to loss of the entire data on your computer's hard disk.

**Linux users.** Just after inserting the card, issue below-mentioned command at the terminal and see what name has

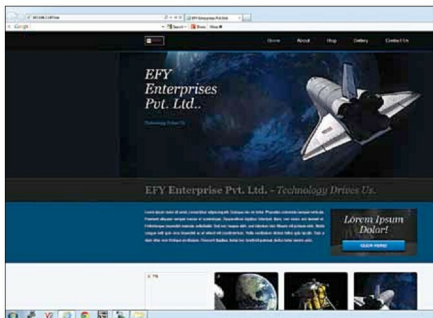


Fig. 1: Sample website template



Fig. 2: Win32 disk imager

been given to the card. Look for the highest order, such as sdc and sdd:

```
ls /dev/sd*
```

The result will be something like mentioned below:

```
efy@Acer-A0D255E:~$ ls /dev/sd*
/dev/sda /dev/sda2 /dev/sda4 /dev/
sda6 /dev/sda1 /dev/sda3 /dev/sda5 /
dev/sda7 /dev/sdb1 /dev/sdb2 /dev/sdc
```

(The SD card has been identified as '/dev/sdc' in the case mentioned

above.)

To double-check, just unplug and re-plug the card a few times and notice the changes.

Now use 'cd' command and go to the folder where you have extracted the image:

```
cd Desktop/Raspberry-Pi/OS
```

where 'Desktop/Raspberry-Pi/OS' is the directory where we extracted the image. This image file now needs to be written on the SD card to make it the OS. At the terminal prompt, issue the following command:

```
sudo dd if=2012-09-18-wheezy-raspbian.img of=/dev/sdc bs=4M
```

When it prompts for 'su' password, supply it to get the image written on the SD card. 'sdc' needs to be replaced with the name your system has given